

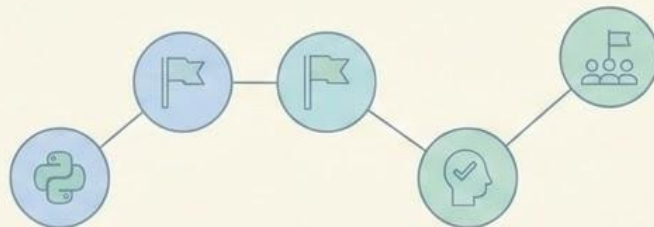
Python for AI Engineering

Mastering The Essential Python Concepts for
Agentic AI System Development



Course Overview

- 5 Sessions × 60 Minutes
- Beginner Friendly
- Hands-on Learning
- Practical Projects



Session 1: Python Fundamentals

Variables, Data Types & Control Flow

- Variables & Types
- If-else conditions
- Loops
- Basic Functions



Session 2: Core Python Deep Dive

{ } Lists & Dictionaries



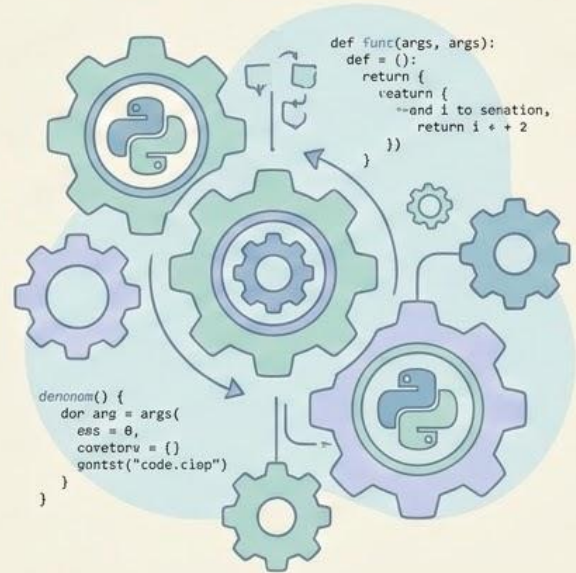
List & Dict Comprehensions



Functions (*args, **kwargs, Decorators)



Generators & Type Hints



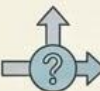
Session 3: Data Structures

Choosing the Right Tool

{ } List, Tuple, Set, Dict

 NamedTuple

 defaultdict, Counter, deque

 When to use which

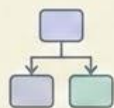
	Lookup Speed	Insertion Speed	Ordered
List			
Tuple			
Set			
Dict			



Session 4: Object-Oriented Python



Classes & Objects



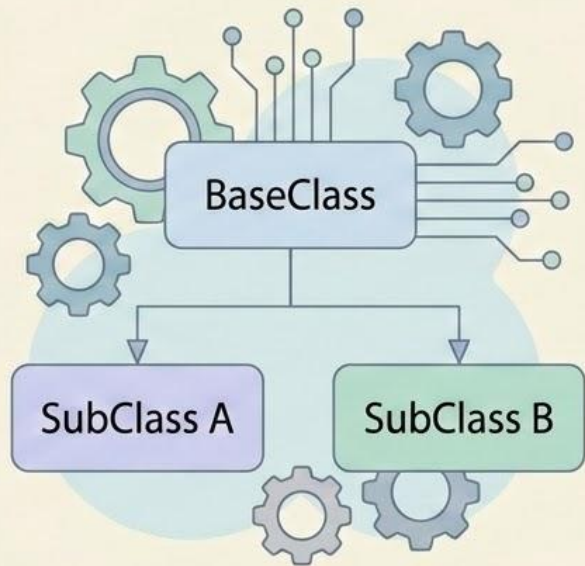
Inheritance &
Polymorphism



DataClasses



Pydantic Models



Session 5: Error Handling & File Operations

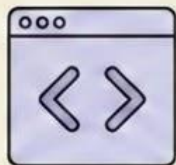
- { } Try, Except & Finally
-  Reading & Writing Files
-  JSON & CSV Handling
-  Mini Project Integration



How We'll Learn



**Live
Coding**



**Hands-on
Exercises**



**Solving
Coding
Challenges**



**Mini
Projects**





**Let's Build Strong Python
Foundations Together**

